

Visualization of Film Circulation and Identifying Patterns of Success

State of the Art Report

Adam Herger 12330620

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Bachelor Thesis

Supervisors: Michaela Tuscher, Velitchko Filipov

Abstract. The following report consolidates existing literature on the topic of visualizing the film festival circuit and identifying patterns of success in large relational datasets. Various papers have previously explored the idea of interpreting film circulation as a network of interconnected nodes. The core problem therefore does not revolve around the question of whether the film festival circuit can be modelled as such, but rather, what constitutes a tool that maximizes the chances of discovering patterns of success. The report consequently suggests the development of an interactive, web-based bipartite force-directed node graph with extensive filtering capabilities that allows for the discovery of emergent properties and thus patterns of success. Furthermore, the report argues that patterns of success cannot simply be defined by discrete numeric values such as gross revenue, festival showcases, or number of awards, but additionally, through consideration of node placement in the graph and surrounding structures, such as clustering and bridging. Finally, the report outlines the gap in literature, and suggests a concrete visualization strategy for the subsequent bachelor thesis.

1 Introduction

Film festivals play an important role in determining the length of a film’s lifecycle and ultimately its success. Festivals provide directors with the opportunity to showcase their films resulting in increased visibility and recognition as well as the chance to receive awards, further boosting the film’s cultural significance. The film festival circuit is especially valuable for smaller, independent films such as short films and documentaries. Thus, it is common practice for film makers and directors to analyze the success of previous similar films to determine an optimal release strategy and what festivals to focus their efforts on.

While large and comprehensive data sets containing detailed information on current film festival circulation already exist, such as the dataset by Skadi Loist and Evgenia Samoilova [LS23], the sheer scale of the dataset makes it incredibly difficult to derive any meaningful conclusions by manual inspection alone. While following a single film’s trajectory across festivals can yield beneficial results and insight for directors, finding more general patterns of success across thousands of different data points is nearly impossible without specialized visualization tools or algorithms.

Thus, the scientific problem I intend to address in my thesis is what constitutes a visualization tool that allows users to discover meaningful patterns of success across the film festival circuit that can help them construct an optimal release strategy for their own film. The core challenge therefore is not simply to visualize thousands of interconnected nodes, but to reveal clusters, trajectories, and structures. Ideally, such a visualization tool would answer questions such as:

- Do groups of festivals exist that are tightly interconnected?
- Do certain festival node paths exist that increase a film’s success rate?
- What festivals are important for a film to be showcased in for certain genres, languages, countries of origin, film type etc. . .
- What notable differences exist between film type in regard to circulation?

Additionally, it is important to define what constitutes patterns of success and what metrics we can measure to discover them. Indicators of a film’s success could include the number of awards a film has won over its lifetime, the number of festivals it was showcased in or its gross total revenue i.e. box-office revenue.

2 Related Work

2.1 Film Festival Circulation

Historically, film research has mostly relied on either individual qualitative case studies or theoretical frameworks as a way to analyze film circulation. However, in recent years, film circulation research has shifted from qualitative research to data driven quantitative research [LS23]. While qualitative research only captures a certain small subset of film circulation, quantitative research allows for a more comprehensive view that captures circulation patterns of thousands of festivals worldwide. Dr. Skadi Loist, Associate Professor of Film Studies in the Department of Art and Media Studies at the Norwegian University of Science and Technology, has been pioneering the quantitative approach to film circulation analysis with their papers [LS23] and [Loi19] outlining a comprehensive relational database of films and festivals obtained primarily through manual data collection and web scraping. This thesis will primarily be exploring and making use of this database for the visualization tool as it is arguably the most comprehensive and structured collection of film festival circulation data. Since the database was built with the FAIR(ER) principles in mind, a framework that defines guidelines intended to optimize the reuse of data for humans and computers, finding, accessing and integrating the data will be a far simpler task and potential issues such as missing values, ambiguous groupings and inconsistent naming conventions can be circumvented.

Zemaityte et al. in their paper *Quantifying the global film festival circuit: Networks, diversity, and public value creation* [ZV24], provide an existing example of a node-based visualization tool that implements a continuous latent vector space system to group festivals instead of discrete labels such as language or production country. While their approach to visualizing the dataset is quite different to the one that will be outlined in this report, it demonstrates that the global film festival circuit can indeed be analyzed as a relational system and reveal meaningful insight, rather than focusing on isolated events. While Zemaityte et al. have already implemented force-directed node graphs for visualizing data sets, the tool itself is primarily used for academic analysis and production of static figures as reference in papers as opposed to an accessible interactive visualization tool for the public use. This thesis will instead focus on a publicly accessible web-based application with interactivity and real-time attribute filtering.

Ehrich, Samoilova, and Loist [Ehr22] explore a different side to film festival circuit analysis. Specifically, how network analysis can reveal structural inequality in the festival sector. In their paper, they model the sector as a one-mode network of festivals as well as a two-mode network of festivals and films and come to the conclusion that films produced by men-only creative teams are statistically more likely to be showcased at festivals than films by women-only teams.

Skadi Loist’s article on queer film circulation [Loi15], while mostly qualitative,

shows that some films circulate through activist festivals and others through art-film festivals, indicating that a film’s circulation is highly dependent on various attributes and it is not prudent to view circulation patterns as homogenous. Additionally, Skadi argues that, at least back in 2015, although debatably still in today’s day and age, society, while portraying itself as having moved into a “post gay” era, still relegates queer centric films to specific LGBTQ activist festivals, while mainstream international art film festivals merely use them as a way of exhibiting artistic or cultural prestige. Skadi’s paper shows that not only can we discover measurable patterns of success through the visualization of festival circuits but also uncover broader concepts such as societal biases.

2.2 Graph Visualization

McGee et al. [MP19] provide a state of the art report on the current research regarding multilayer network visualization, showcasing the differences between multiplex, multimodal, and multilayer representations. Although most of the systems outlined in the paper such as matrix displays and hybrid views exceed the scope of this thesis, it still serves as a valuable resource to determine how a node based visualization graph should be constructed.

Kobourov’s paper [Kob24] on spring embedders and force directed graph drawing algorithms outlines its inherent benefits and drawbacks. Force directed graphs are flexible, aesthetically pleasing and very effective at revealing clusters and groupings, which is perfect for the purposes of discovering patterns of success in film circuits. However, the paper also talks about the scalability issue of calculating physics simulations for a list of growing nodes in such a graph.

Tong et al. and their paper [TFGER07] address the issue of finding meaningful sub-structures in large node graphs without relying on exact query matches. They propose a method designed for graphs with nodes containing attributes, which returns not only exact query matches, but also near matches depending on several parameters. This could serve as a useful feature for discovering circulation patterns in the film festival circuit as some users may be more interested in researching similar circulation patterns to their chosen criteria rather than forcing exact matches, which may inhibit the number of useful sub structures as a result.

Ortega Mattsson’s paper on comparing readability of the force-directed and orthogonal graph layout [OM20] supports the idea of implementing force directed visualization techniques to visualize the film festival circuit. Following a user study in which participants are tasked with completing topology-based tasks using both force directed and orthogonal layouts, the paper concludes that overall, the force directed node graph layout was more advantageous for participants.

Ganser and North and their paper [GN98] while fairly dated still provide a foundational overview of the benefits and limitations of using force directed layouts for visualization. The authors argue that force directed layouts specialize

in revealing symmetries, structures, and clustering, however, also state that the nodes themselves must be kept simple, as adding in additional labels or descriptions decrease readability.

Holten and van Wijk's paper on force directed edge bundling for graph visualization [HvW09] addresses the problem of distinction between edges when nodes become clustered. The paper suggests edge bundling as a way of combining close edges and thus reducing unnecessary clutter while not reducing the graphs usefulness.

3 Method

The literature presented in the previous chapter was compiled around the core thesis topic of finding patterns of success in the relational film circulation database. The purpose of the literature review was to assemble a set of papers that together explain the two main ideas surrounding this thesis, namely the film festival circuit and finding patterns of success, and the visualization techniques i.e. primarily bipartite force directed node graphs.

The search strategy revolved around finding relatively new papers, ideally less than ten years old, using keywords such as “Film Festival Circulation”, “Finding Patterns of Success in the Film Festival Circuit”, and “Bipartite Force Directed Node Graphs”. Additionally, the papers had to contribute a unique viewpoint or technique as to not have duplicate papers with similar information. Furthermore, in regard to film circulation studies, the methodology had to be mainly quantitative as opposed to qualitative. Regarding the graph visualization portion of the papers, they had to mostly discuss the general framework and methodology of bipartite force directed node graphs, as the final visualization tool will be built using React and the D3.js library as opposed to programming one completely from scratch. Therefore, more math oriented papers were not considered for the purposes of this thesis. Most of the papers were found in open source journals and repositories such as Springer, NECSUS, arXiv, PLOS ONE, Applied Network Science, and ResearchGate.

As aforementioned, the resulting set of papers was compiled as to present a broader bird’s eye view of the thesis statement and topic, as opposed to going into depth into a certain area.

4 Results

The above mentioned literature points to the idea of film circulation being best modeled as a bipartite network of nodes, each containing a list of attributes. This thesis will therefore focus on building a visualization tool that combines the emergent properties of a relational graph view with selective filtering features to limit the complexity of the resulting graph. To emphasize, the data must first be modeled as a network of films and festivals, second, the visualization needs to expose emergent properties such as clusters, bridges and prominent nodes, and third, it must allow users to filter down the resulting set of nodes by attributes, so as to not overwhelm them with a large, interconnected web of thousands of nodes and links.

4.1 Quantitative Film Circulation Analysis

Loist and Samoilova [LS23] and Zemaityte et al. [ZV24] make up the basis of this thesis as, for one, they show that the film festival circuit can indeed be analyzed through quantitative means, as well as visualized using bipartite force directed node graphs. An ideal visualization tool must model festivals and films as nodes containing filterable attributes.

4.2 Patterns of Success and Emergent Properties

The literature suggests several ways of characterizing patterns of success. One option is to consider the number of festivals a film was showcased in as this indicates widespread appeal across critics and fans. Another way would be to measure the prestige of a film, for example the number of awards it has won or the exclusivity of the festivals it was showcased in. Additionally, considering the gross revenue of a film is a simple yet powerful metric to evaluate success. However, Loist argues that the film festival circuit differs substantially to more commercial oriented releases and cannot be primarily measured using the same metrics of financial achievement [Loi15]. Furthermore, some films may not have been financially successful or haven't been shown at a large number of festivals but could still be considered successes due to being influential in their respective sub-circle.

Another important factor to consider when evaluating patterns of success is emergent properties. This includes node clustering, link abundance and bridges. Metrics that, rather than measuring a single defined numeric parameter, reveal themselves through the physical interaction between nodes and links. The influence of a film may not be strictly tied to discrete values such as prominence or revenue, but also to its influence as a node in such a force directed graph. For example, a node that bridges two mostly separate portions of a graph may very well be categorized as influential, while not exhibiting the common attributes of an influential film. This supports the idea that allowing users to toggle between various different metrics of success depending on their needs is the optimal path

to achieving a flexible visualization tool, aside from simply discarding the idea of discrete attributes and metrics and instead implementing a latent vector space system as outlined by Zemaityte et al. in their paper [ZV24].

4.3 Visualization Models

The literature does not present a single definite best way of modelling the film festival circuit. Most papers implement a node link graph, however, there is no clear consensus on whether using discrete attributes, programming complex layered views and hybrid systems [MP19], or discarding the idea of attributes entirely and instead relying on latent vector space [ZV24] is the best option. For the purposes of this thesis a simple bipartite force directed node graph will suffice as a prototype. The important aspect is implementing the force directed layout, as this allows for users to discover emergent properties, which, as previously mentioned, can reveal important patterns of success.

Additionally, Kobourov makes it clear that a well structured filtering system is crucial as to not overload the user with redundant or irrelevant information and to allow the user to focus on the precise metrics they require [Kob24].

Furthermore, [MP19] and [YT15] explore how multimodality can be used to create layers of information as to further separate information and make it more manageable for the user. While this is beyond the scope of this thesis, it may be an interesting feature to explore in future research.

4.4 The Database

For the purposes of this thesis the visualization tool will be powered by the film circulation dataset compiled by Skadi Loist and Evgenia Samoilova [LS23]. The database is comprised of four core components: the film dataset containing a set of over 9000 unique films, a survey dataset covering survey questions, an IMDb dataset comprising awards, companies, crew, festival runs and film metadata, and a festival dataset which outlines information such as festival name, location, award recognition, and specialization.

The database is incredibly comprehensive and must thus be cleaned i.e. pre-processed before integrating it into the web application. Many variables appear redundant, relevant in only extremely niche circumstances, or are not useful for discovering patterns of success. While the final set of attributes included in the visualization tool is not final, the following table shows likely candidates for the most important attributes to consider.

Dataset	Attributes
Film	unique.id sample.year prod.year length.min prod.country regions genre
Survey	Screenings Budget Marketing Costs
IMDb	imdb.awards imdb.crew imdb.festival_runs imdb.release_info
Festival	festival.id fest.name fest.location festival.categories

Additionally, unique identifiers and foreign keys will be useful to draw relations between films, festivals, and their respective surveys and IMDb pages. Due to the sheer number of attributes included in the dataset, in order to achieve a more user-oriented interface, only certain “high value” attributes should be exposed to the user. Hidden attributes may be toggled via an advanced mode settings for those who wish to explore the dataset in more detail.

5 Discussion

The existing literature already showcases that presenting the film festival circuit as a series of interconnected nodes is not only possible, but, generally considered to be the most practical way of discovering valuable metrics and emergent properties i.e. patterns of success. The thesis will therefore instead focus on how such a visualization tool can be optimized. What does an efficient data to graph view pipeline look like? What features can be implemented to prevent information overload for the user and provide filtering capabilities to focus on certain attributes or sub-structures? What constitutes a pattern of success and how can the tool be optimized to reveal them?

One major advantage of the proposed single layer graph view is that, while perhaps more limited than the latent vector space graph proposed by Zemaityte et al. [ZV24], it is far more intuitive which is crucial for a user oriented visualization tool. Users can easily view the dataset in its entirety to discover emergent properties, then filter down the result to inspect clusters and sub-structures in more detail. Additionally, the proposed tool updates its view in real time, as opposed to some of the other models discussed earlier in the literature which are primarily static figure generators.

However, there are some important factors to consider that if left ignored could impact the effectiveness of the visualization tool. For one, if the dataset is too large, even with attribute filtering it may be difficult for users to narrow down their search and discover emergent properties. Additionally, since the tool relies on discrete values in the form of attributes as opposed to a more fuzzy logic oriented approach as previously mentioned in Zemaityte et al.’s paper [ZV24], some emergent properties that don’t share explicit discrete value similarities may be lost. Furthermore, since the proposed graph view is real time and interactive, depending on the size of the dataset, there may be performance and responsiveness issues on the user end.

To emphasize, the existing literature has gaps when it comes to publicly deployed, interactive, user-oriented visualization tools. This gap is precisely what the eventual bachelor thesis will attempt to fill.

6 Conclusion

The current literature indicates that the film festival circuit can not only be analyzed quantitatively using relational databases, but also that force directed node graphs are arguably the most effective way of visualizing this data with user interactivity, emergent properties, and discovery of patterns of success in mind. Additionally, the main challenges across the set of literature remains scalability as well as mitigations against information overflow.

For the proposed bachelor thesis, the ideal path to achieving such a visualization tool is building a bipartite force-directed interactive node graph web application with extensive filtering capabilities, with React and D3.js serving as the framework. Such a tool will fill the gap in the existing literature and ideally serve as a meaningful contribution to further film festival circuit research.

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